

Minor Project 2

Screenshots

1. Search (Products tab)

Searching by keyword returns matching records from the Name, SKU, and Category fields.

The screenshot displays a web application window titled "Warehouse Inventory & Order Manager". The interface includes a navigation bar with tabs for "Products (Search & CRUD)", "Orders (CRUD)", "Administration (Suppliers)", and "Settings". The "Products (Search & CRUD)" tab is active. Below the navigation bar, there is a search section with a text input field containing "IND-201" and a "Clear Search" button. Below the search field, there are input fields for "ID: 5", "SKU: IND-201", "Name: Heavy Duty Pallet Stra", and "Category: Warehouse". There are also input fields for "Price: 14.50", "Stock: 250", and a "Supplier" dropdown menu showing "Apex Global Logisti". To the right of these fields are three buttons: "Insert" (green), "Update" (blue), and "Delete" (red). Below the search section is a table with the following columns: ProductID, SKU, ProductName, Category, UnitPrice, StockQuantity, and SupplierID. The table contains one row with the following data: ProductID 5, SKU IND-201, ProductName Heavy Duty Pallet Stra, Category Warehouse, UnitPrice 14.50, StockQuantity 250, and SupplierID 1. The table is set against a blue background.

ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
5	IND-201	Heavy Duty Pallet Stra	Warehouse	14.50	250	1

Figure 1a: Search term matching the SKU field

Warehouse Inventory & Order Manager

Products (Search & CRUD)

Orders (CRUD)

Administration (Suppliers)

Settings

Search (Name, SKU, Category): Furniture

Clear Search

ID: 3

SKU: OFFC-101

Name: Ergonomic Desk Chair

Category: Furniture

Price: 199.50

Stock: 18

Supplier: Midwest Industrial S

Insert

Update

Delete

	ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
▶	3	OFFC-101	Ergonomic Desk ...	Furniture	199.50	18	3
	4	OFFC-102	Adjustable Stand...	Furniture	349.00	10	3

Figure 1b: Search term matching the Name or Category field

2. Insert

Warehouse Inventory & Order Manager

Products (Search & CRUD)
Orders (CRUD)
Administration (Suppliers)
Settings

Search (Name, SKU, Category):

ID:
SKU:
Name:
Category:

Price:
Stock:
Supplier:

	ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
▶	1	ELEC-001	Wireless Optical ...	Electronics	24.99	120	2
	2	ELEC-002	Mechanical Gami...	Electronics	89.99	45	2
	3	OFFC-101	Ergonomic Desk ...	Furniture	199.50	18	3
	4	OFFC-102	Adjustable Stand...	Furniture	349.00	10	3
	5	IND-201	Heavy Duty Palle...	Warehouse	14.50	250	1
	6	IND-202	Hydraulic Hand T...	Warehouse	285.00	8	1

Figure 2a: New record entered, before insert

Warehouse Inventory & Order Manager

Products (Search & CRUD)

Orders (CRUD)

Administration (Suppliers)

Settings

Search (Name, SKU, Category):

Clear Search

ID: 8

SKU: ELEC-045

Name: Cordless Headset

Category: Electronics

Price: 149.99

Stock: 20

Supplier: Pacific Tech Compoi

Insert

Update

Delete

	ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
	1	ELEC-001	Wireless Optical ...	Electronics	24.99	120	2
	2	ELEC-002	Mechanical Gami...	Electronics	89.99	45	2
	3	OFFC-101	Ergonomic Desk ...	Furniture	199.50	18	3
	4	OFFC-102	Adjustable Stand...	Furniture	349.00	10	3
	5	IND-201	Heavy Duty Palle...	Warehouse	14.50	250	1
	6	IND-202	Hydraulic Hand T...	Warehouse	285.00	8	1
▶	8	ELEC-045	Cordless Headset	Electronics	149.99	20	2

Figure 2b: New record appears in the grid

3. Update

Warehouse Inventory & Order Manager

Products (Search & CRUD) Orders (CRUD) Administration (Suppliers) Settings

Search (Name, SKU, Category): Clear Search

ID: SKU: Name: Category:

Price: Stock: Supplier:

	ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
	1	ELEC-001	Wireless Optical ...	Electronics	24.99	120	2
	2	ELEC-002	Mechanical Gami...	Electronics	89.99	45	2
	3	OFFC-101	Ergonomic Desk ...	Furniture	199.50	18	3
	4	OFFC-102	Adjustable Stand...	Furniture	349.00	10	3
	5	IND-201	Heavy Duty Palle...	Warehouse	14.50	250	1
	6	IND-202	Hydraulic Hand T...	Warehouse	285.00	8	1
▶	8	ELEC-045	Cordless Headset	Electronics	149.99	20	2

Figure 3a: Existing record edited, before update

Products (Search & CRUD) Orders (CRUD) Administration (Suppliers) Settings

Search (Name, SKU, Category): Clear Search

ID: SKU: Name: Category:

Price: Stock: Supplier:

	ProductID	SKU	ProductName	Category	UnitPrice	StockQuantity	SupplierID
	1	ELEC-001	Wireless Optical ...	Electronics	24.99	120	2
	2	ELEC-002	Mechanical Gami...	Electronics	89.99	45	2
	3	OFFC-101	Ergonomic Desk ...	Furniture	199.50	18	3
	4	OFFC-102	Adjustable Stand...	Furniture	349.00	10	3
	5	IND-201	Heavy Duty Palle...	Warehouse	14.50	250	1
	6	IND-202	Hydraulic Hand T...	Warehouse	285.00	8	1
▶	8	ELEC-045	Cordless Headset	Electronics	90.99	20	2

Figure 3b: Updated record in the grid

4. Delete

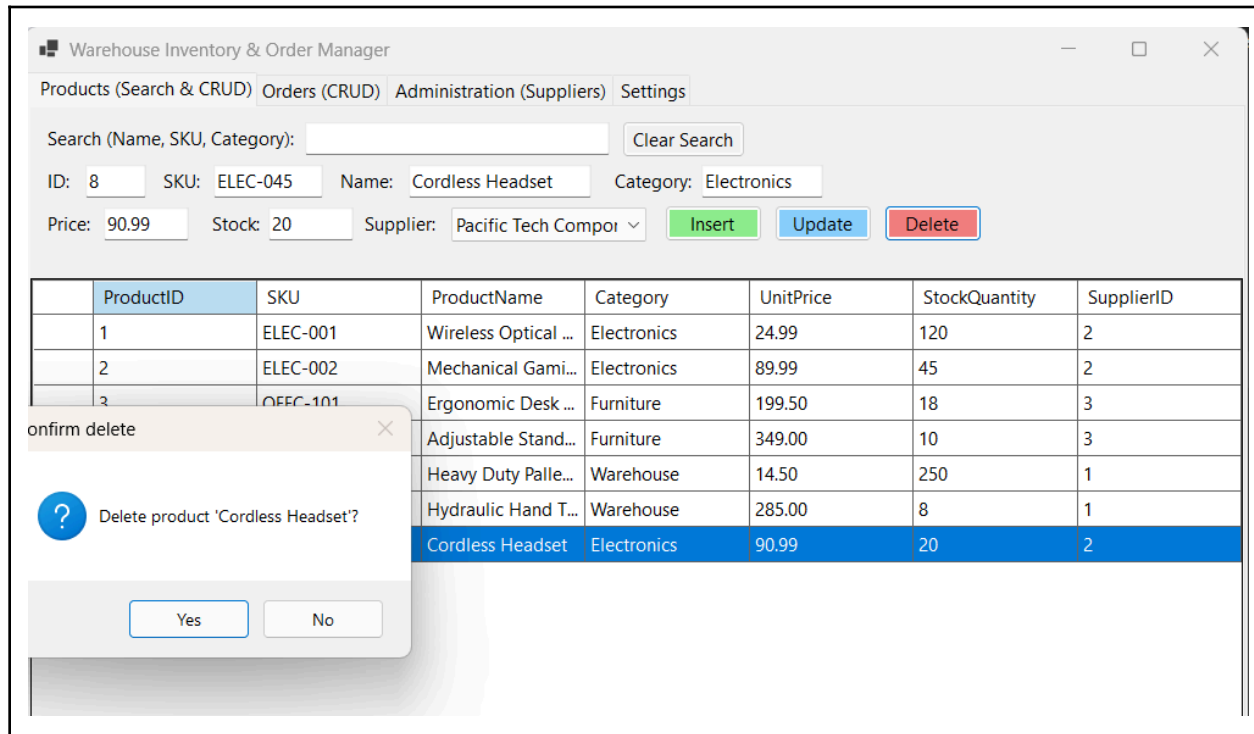


Figure 4a: Delete confirmation

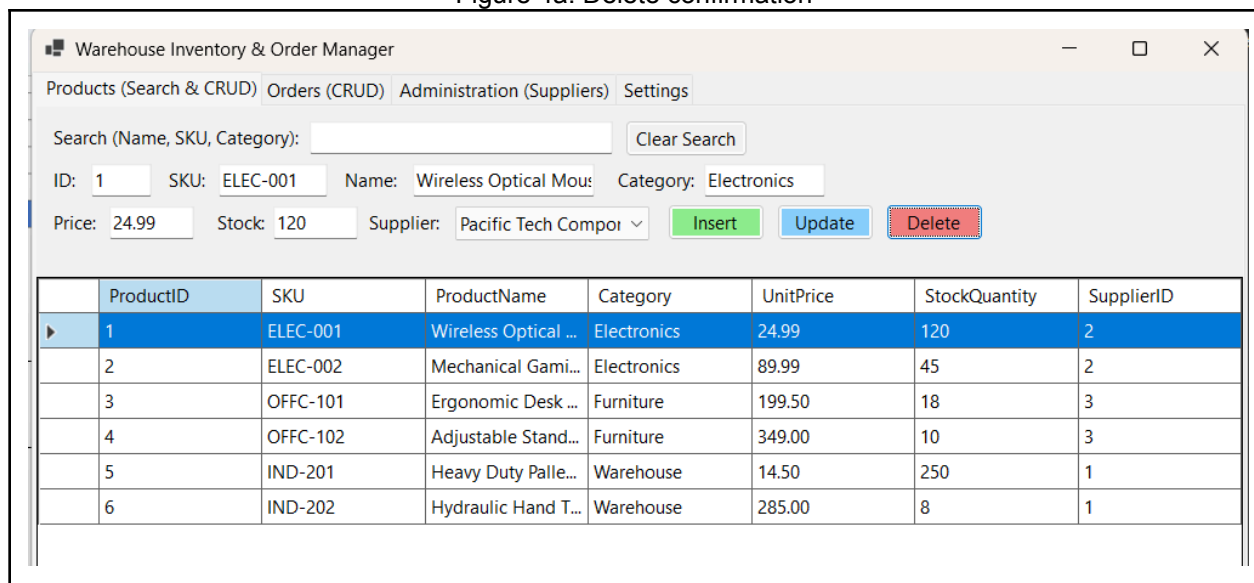


Figure 4b: Record removed from the grid

Additional Screenshots (optional)

Warehouse Inventory & Order Manager

Products (Search & CRUD) Orders (CRUD) Administration (Suppliers) Settings

☒ Enable Admin Mode (Allow altering Suppliers table)

ID: Company: Email: Phone: City:

	SupplierID	SupplierName	ContactEmail	Phone	City
▶	1	Apex Global Logistics	contact@apexlogistics.co...	555-0101	Atlanta
	2	Pacific Tech Components	sales@pacifictech.com	555-0102	Seattle
	3	Midwest Industrial Supply	orders@midwestind.com	555-0103	Chicago

Figure 5: Admin-only Suppliers management, (enable with checkbox)

Warehouse Inventory & Order Manager

Products (Search & CRUD) Orders (CRUD) Administration (Suppliers) Settings

Application Theme:

Figure 6: Dark mode setting applied

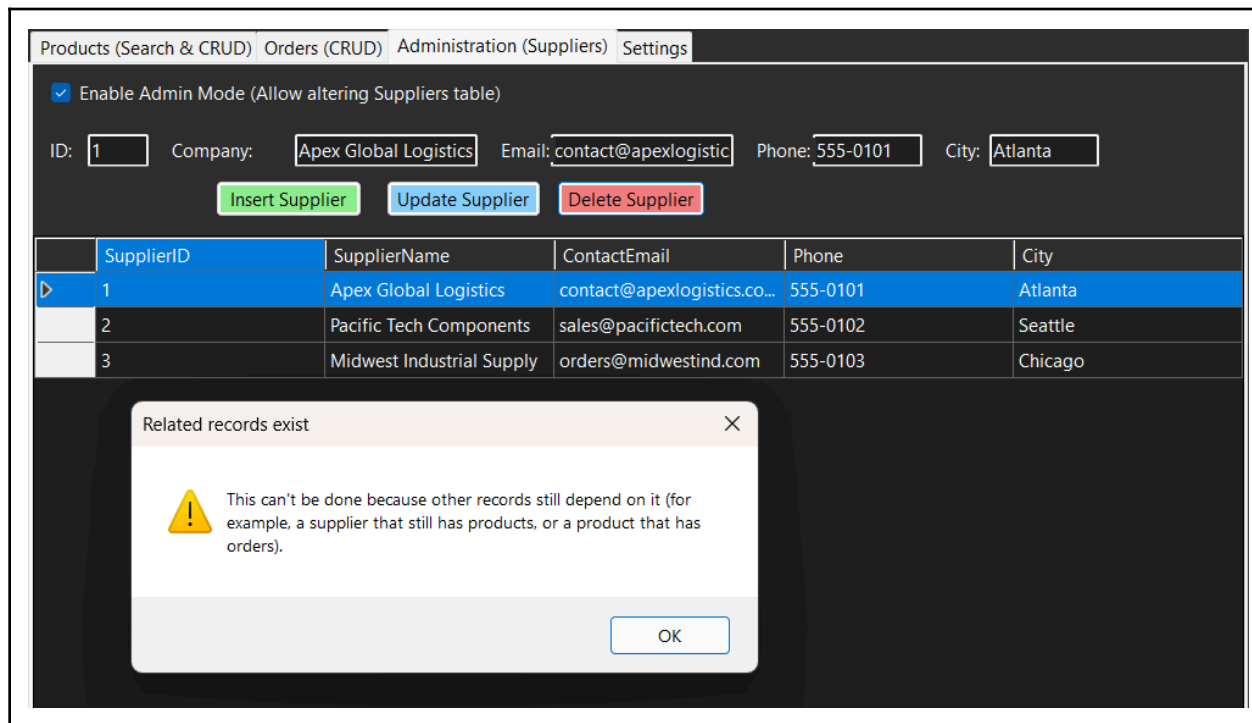


Figure 7: Foreign key protection message

Written Responses

1. Describe a challenge you overcame in this project

The biggest challenge was handling the relationships between tables safely. My first draft of the database used ON DELETE CASCADE on the foreign keys, which meant deleting a single supplier and silently deleting all of its products and every order for those products. I removed that so SQL Server blocks the delete instead. In the application I catch the foreign key error (SQL error 547) and show a clear message explaining that other records still depend on the item, and I added a Yes/No confirmation before every delete, to prevent accidental deletes.

Another smaller challenge was the layout: labels overlapped their text boxes, which I fixed with auto sizing labels and font-based form scaling.

2. Describe a technique you learned in this project

For the UI, I set up DisplayMember and ValueMember on ComboBoxes so users can pick suppliers and products by name rather than manually entering IDs. I also centralized all database calls into a single data layer class so the forms weren't directly cluttering up the queries.

3. What more would you want to refine to make it ready to sell, ship, and deploy?

I would like to add real authentication and role-based permissions in place of the Admin Mode checkbox, to make it more legitimate and secure.

4. Which technology did you choose (WPF, WinForms, WinUI) and why?

I chose WinForms. It is the quickest to build for a data-entry application: the event model is simple, there is no XAML or MVVM pattern, and controls bind directly to database results.